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A spatiotemporal prediction model for light pollution in conservation areas using remote sensing datasets

Lala Septem Riza [a], Zulfikar Ali Yunara Putra [a], Muhammad Fajar Yusuf Firdaus [a], Fajar Zuliansyah Trihutama [a], Ahmad Izzuddin [a], Judhistira Aria Utama [b,*], Khyrina Airin Fariza Abu Samah [c], Dhani Herdiwijaya [d], Rinto Anugraha NQZ [e], Emanuel Sungging Mumpuni [f]

[a] Department of Computer Science Education, Universitas Pendidikan Indonesia, Bandung 40154, Indonesia
[b] Department of Physics Education, Universitas Pendidikan Indonesia, Bandung 40154, Indonesia
[c] College of Computing, Informatics and Media, Universiti Teknologi MARA Cawangan Melaka Kampus Jasin, Melaka City 77300, Malaysia
[d] Astronomy Research Division, Institut Teknologi Bandung, Bandung 40132, Indonesia
[e] Faculty of Mathematics and Natural Sciences, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia
[f] National Research and Innovation Agency, Jakarta 10340, Indonesia

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ABSTRACT

Artificial Light at Night (ALAN) significantly threatens protected areas from urbanization. As urbanization continues to grow, there is a need for forecasting future light pollution and ALAN for the protected areas in Indonesia. This study proposes a four-step computational model for forecasting spatial–temporal light pollution in nine protected areas in Indonesia via spatiotemporal modeling and linear models. The four steps for predicting spatial–temporal light pollution are (i) data collection, (ii) data pre-processing, (iii) model and prediction of population, and (iv) model and prediction of light pollution. Two critical data must be provided: population data from the review area and light pollution data generated by the Earth Observations Group (EOG) at the National Oceanic and Atmospheric Administration (NOAA)/National Centers for Environmental Information (NCEI). We are using nine conservation areas in Indonesia, including the Kuningan Botanical Gardens, Bosscha Observatory, Timau Kupang National Observatory, Sermo Reservoir, Mount Batur Geopark, Sewu Mountains Geopark, Mount Rinjani Geopark, Lake Toba Geopark, and Belitong Geopark. The developed model involves a linear model to predict ALAN with spatial–temporal modeling. We present long-term predictions for the next 20 years.

1. Introduction

A critical component for the protection of natural and cultural heritage comes in the form of protected areas. Historically, protected areas emerged due to concern over human-made environmental changes and the loss of wilderness [1]. They served as a refuge for wildlife and natural features [2]. In addition, conservation areas protect culturally important places such as many different areas in Egypt [3] or Aboriginal cultural heritage in Tasmania [4]. Today, they carry much importance in a growing number of aspects. Conservation areas aim to preserve the natural and cultural heritage and serve economic, social, and other conservation objectives [5]. There are 253,408 protected areas in 245 countries and territories, covering 15.79% of terrestrial and inland waters [6]. In addition, 17,786 protected areas are marine and cover 8.15% of the oceans. In the long term, these protected marine areas potentially benefit livelihoods that depend on fisheries [7]. On land, protected areas also improve the livelihoods of local communities, especially in conjunction with tourism [8,9].

Conservation itself is one side of sustainability. As outlined in the United Nations Sustainable Development Goals (SDGs) at [5], which include part of their historical purpose of wildlife stock, but also water [10], food security such as through wild relatives of essential crops Danny [11] and fishery reserves [12,13], and carbon sequestration [1] among many others.

In 2001, International Darksky Association (IDA) established the International Dark Sky Places (IDSP) conservation program as a way to encourage society to use responsible lighting and educate the public.

* Corresponding author.
E-mail addresses: lala.s.riza@upi.edu (L.S. Riza), zulfikaryunara@upi.edu (Z.A.Y. Putra), mfajaryusuf@upi.edu (M.F.Y. Firdaus), fajarzuliansyaht@upi.edu (F.Z. Trihutama), ahmadizzuddin@upi.edu (A. Izzuddin), j.aria.utama@upi.edu (J.A. Utama), khyrina783@uitm.edu.my (K.A.F.A. Samah), dhani.herdiwijaya@gmail.com (D. Herdiwijaya), rinto@ugm.ac.id (R.A. NQZ), emanuel.sungging.mumpuni@brin.go.id (E.S. Mumpuni).

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Dark sky tourism can help mitigate this detrimental impact and further serve the SDGs [14]. Furthermore, spending time in dark sky areas promotes health, well-being and connectedness with nature [15].

Protected areas, however, are under attack from multiple dangers. One of the more widely recognized dangers is from poachers and criminal groups who are after species with high monetary value, such as tigers [16]. Although the probability was higher at lower elevations close to roads and tourist trails, it can be mitigated by ranger patrols and new ranger posts. Another danger is the illegal logging of trees on protected sites [17]. This activity endangers the species of trees being harvested and the organisms that rely on them for their habitat [18]. Illegal logging also allows easier access deeper into the protected area for poachers. Logging also negatively affects protected areas' carbon sequestration [19]. After logging, the threat of livestock grazing in protected areas also comes, which can impact the wildlife in the areas as mentioned earlier [20]. An emerging threat comes from the development in and around protected areas. As cities and settlements urbanize, as much as 8% of the terrestrial vertebrate species on the International Union for Conservation of Nature (IUCN) Red List come under threat [21]. Trends suggest that urbanization threatens to lead to local but significant biodiversity degradation. Industrial activity in the surrounding areas also directly affects a number of fish species in the protected area with bioaccumulation of pharmaceuticals, illicit drugs and their metabolites due to water systems. inland receiving wastewater. [22].

Artificial Light at Night (ALAN) is one very important threat to protected areas from urbanization. There is a continuously growing body of evidence on the ecological impacts of light pollution [23]. For avians, this can mean many disruptions starting from song timing [24] to restless rest, which could lead to more parasites [25]. Birds also become navigationally disoriented, with seabird fledglings grounded on roads and beaches near colonies [26] and nocturnally migrating birds are disoriented by ALAN [27]. Marine animals do not escape the problem of light pollution either; turtles' navigation becomes disrupted due to skyglow and lights from industrial areas [28]. Several works highlight ALAN's impact on protected areas and conservation efforts ([29,30]). Specifically, in the effort of firefly conservation, while protected areas have been able to buffer the effects of urbanization and deforestation, protected areas have been found to be incapable of stopping light pollution from disrupting firefly reproduction [31]. Because of the growing awareness of light pollution, especially near protected areas, further examination and sound policies must be created [32]. In aid of the policymaking process, previous works have also approached the issue of spatiotemporal modeling of light pollution in protected areas. Xiang and Tan [33] modeled the changes in light pollution, showing a significant increase in light pollution in Chinese protected areas. They found that there were regional differences in the changes in light pollution. Mu et al. [34] evaluated the status of light pollution in global protected areas by classifying them into non-polluted, continuously polluted, and discontinuously polluted. They found that light pollution in protected areas is growing over time except in Japan and the USA. These studies help understand the current situation, which could lead to better policy. However, as the pace of urbanization continues to grow in Indonesia [35,36], there is a need for forecasting future light pollution and ALAN for the protected areas in Indonesia. As a broad interdisciplinary subject, there are many approaches to studying light pollution, including the research objective, data collection, pre-processing, and data analysis [37].

Therefore, in this research, we are aiming to forecast light pollution in nine protected areas in Indonesia via spatiotemporal modeling using linear models based on population. Our choice of the linear model for spatiotemporal modeling is partly due to its simplicity and explainability. This explainability component is essential for building trust with stakeholders [38]. We use population as the principal explanatory variable due to its applicability in this context [39] and its implications for city planning [40]. This study will enable stakeholders to assess the implications of future light pollution with a case study of nine protected areas in Indonesia. Moreover, we use light pollution generated by the Earth Observations Group (EOG) at the National Oceanic and Atmospheric Administration (NOAA)/National Centers for Environmental Information (NCEI), obtaining daily mosaic images and monthly composites using night-time data from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB). VIIRS DNB sensors are on the Suomi National Polar-orbiting Partnership (NPP) satellite and on the NOAA-20 satellite [41].

2. Research material and methods

This section will discuss the data to be used as case studies or experiments, the computational model to obtain long-term predictions, and experimental scenarios. Detailed information on each section is explained in the following subsections.

2.1. Data for case study

The conservation area data used as research objects are based on the issue of conservation area sustainability which is influenced by many factors, one of which is light pollution around the area. In this study, we consider conservation areas consisting of botanical gardens, observatories, and geoparks, also known as earth parks [42]. As shown in Fig. 1, The data covers 9 conservation areas spread throughout Indonesia, as follows:

(a) Kuningan Botanical Garden
Kuningan Botanical Garden is a conservation area in Kuningan Regency, West Java. The area is about 156 hectares and is directly adjacent to Mount Ciremai National Park. The Kuningan Botanical Gardens was built in 2014. It was closed in 2016 due to construction and reopened in 2018. This location contains hundreds of diverse plant species. As home to a lot of vegetation, the issue of light pollution must be considered because of its potential impact on organisms [43,44]. The bounding box coordinates of the site are 7.2° S and 108.383° E to 6.783° S and 108.766° E, with the center point at 6.98° S and 108.40° E.

(b) Bosscha Observatory
Bosscha Observatory is one of the oldest astronomical observatories in Asia. Bosscha is located in Lembang, West Java. Bosscha. It was built in 1923 and was completed 5 years later in 1928. The Dutch Indies Astronomical Association or Nederlandsch-Indische Sterrenkundige Vereniging (NISV), which started the construction of Bosscha, handed over this observatory to the government of the Republic of Indonesia on October 17, 1951. Today, the observatory is a center for astronomical activities such as stargazing and research. Due to the sensitivity of the activities conducted at Bosscha to the state of light pollution around this area, it is urgent to assess the future state of light pollution in the area. The bounding box coordinates of the site are 7.21° S and 107.17° E to 6.37° S and 108.07° E, with the center point at 6.82° S and 107.62° E.

(c) Timau National Observatory, Kupang
Timau National Observatory is a protected area located in Kupang, East Nusa Tenggara. To be precise, it is located in a protected forest area on Mount Timau at an altitude of about 1300 m above sea level. This area is adjacent to Bitobe Village, which is included in the Central Amfoang District. The Observatory at Mount Timau is the first Dark Sky National Park in Indonesia and the largest observatory in Southeast Asia. Therefore, this area was chosen as a case study to ensure that there is no excessive light pollution in the area so that the use of this area as a tourist spot for dark skies and an observatory is optimal. The bounding box coordinates of the site are 10.05° S and 123.49° E to 9.16° S and 124.40° E, with the center point at 9.60° S and 123.95° E.

Fig. 1. Distribution map of the conservation areas used as case studies in this study, namely nine sites in Indonesia.

(d) Waduk Sermo
Waduk Sermo or Sermo Dam is a reservoir located in Kulonprogo Regency, Yogyakarta Special Region, Indonesia. This reservoir was built in 1994 and inaugurated by President Soeharto on November 20, 1996. The location of the reservoir is used as the object of academic discussion regarding the evaluation of geology and the risk of soil movements around the reservoir. The bounding box coordinates of the site are 7.97° S and 109.98° E to 7.69° S and 110.26° E, with the center point at 7.83° S and 110.12° E.

(e) Gunung Batur Geopark
Gunung Batur Geopark is an earth park located in Mount Batur, Kintamani District, Bangli Regency, Bali. The landscape in this area has several unique features, such as a very large caldera and a crescent-shaped lake. Meanwhile, its archaeological uniqueness lies in the various relics that are unique and typical of the Balinese area found on Mount Batur. The biodiversity of the area is also significant. With its diverse natural wealth, assessment of light pollution in the area is vital so that it can be adequately controlled for the sustainability of Gunung Batur Geopark. The bounding box coordinates of the site are 8.42° S and 115.23° E to 8.15° S and 115.50° E with the center point at 8.28° S and 115.37° E.

(f) Sewu Mountain Geopark
The Sewu Mountains Geopark is an earth park area in the Southern Mountains of East Java. The United Nations Educational, Scientific and Cultural Organization (UNESCO) recognized this geopark in September 2015. The Sewu Mountains Geopark extends from east to west for 120 km. The north is bordered by the active volcanoes Merapi and Lawu, while the Indian Ocean borders the south. This area is located at an altitude between 5 m to 700 m above sea level with thousands of limestone hills. Two ecological systems inhabit this area as endokarst and exokarst. Both have different characteristics of biodiversity. To maintain this diverse ecosystem, assessment of light pollution in the area is important so that the sustainable development of the conservation area can be achieved. The bounding box coordinates of the site are 8.17° S and 110.36° E to 7.90° S and 110.63° E, with the center point at 8.00° S and 110.50° E.

(g) Rinjani Geopark
Mount Rinjani Geopark is a conservation area located on the island of Lombok in Southeast Asia. Mount Rinjani Geopark has a rich and diverse landscape with various types of forests ranging from savanna and semi-deciduous forests to low-mountain green forests and tropical mountain green forests. The volcanic complex developed due to the subduction of the Indian Ocean Plate under the Southeast Asian Plate. Mount Rinjani is currently the highest peak in the volcanic complex. Mount Rinjani formed about 12,000 to 6000 years ago. This area is used as a research site and a source of germplasm collection. The bounding box coordinates of the site are 8.51° S and 116.38° E to 8.24° S and 116.65° E, with the center point at 8.37° S and 116.50° E.

(h) Toba Lake Geopark
The Lake Toba Geopark is a conservation area that UNESCO has recognized in 2020. It is located on the island of Sumatra, precisely in North Sumatra Province. This area includes Lake Toba and several other geographical sites. Lake Toba formed as a result of a mega-volcanic eruption 74,000 years ago. This mega-eruption collapsed the roof of the Toba magma chamber and exposed the bedrock in this area. The Toba Caldera covers an area of 1130 square kilometers and is filled with rainwater which supplies the lake. There is a lot of biodiversity in the Toba area or Samosir Island. The bounding box coordinates of the site are 2.44° N and 98.51° E to 2.71° N and 98.79° E with the center point at 2.58° N and 98.65° E.